

CodeAlpha Cyber Security Internship

TASK 4 — NETWORK INTRUSION DETECTION SYSTEM

Suricata-Based Network IDS — Configuration, Detection, Alerts & Response

Project scope: This is a controlled, educational IDS project designed for authorized/lab traffic. It should be deployed only on networks you own or are explicitly authorized to monitor.

1. Objective

The goal is to design a network-based Intrusion Detection System (IDS) using Suricata. The project covers IDS setup, detection rules and alerts, continuous monitoring, basic response procedures, and an optional dashboard/graphing approach. These are the activities specified in the CodeAlpha Task 4 brief.

2. Architecture

Network traffic → Network interface / traffic source → Suricata IDS → Detection rules → Alerts/logs → Analyst review → Response actions → Optional dashboard/graphs

3. Tools and Components

- Suricata — network threat detection engine
- Rules/signatures — identify suspicious traffic patterns
- EVE JSON / alert logs — structured security events
- Command-line tools — configuration and log inspection
- Optional visualization — a dashboard or graphing tool can consume alert logs

4. Example Lab Setup

A safe lab can use a dedicated test machine or virtual network. Suricata monitors a selected interface and writes alerts to its log directory. Testing should use benign, controlled traffic rather than unauthorized scanning or attacks.

Example configuration concepts

```
# Example interface setting in suricata.yaml
HOME_NET: "[192.168.1.0/24]"
# Monitor the authorized lab interface
# interface: eth0
# Enable EVE JSON alerts/logging
```

5. Detection Rules

Rules should detect clearly defined suspicious patterns and generate alerts that an analyst can investigate. The examples below are intentionally simple and suitable for a controlled lab demonstration.

```
alert icmp any any -> $HOME_NET any (msg:"LAB ICMP Detection"; itype:8; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET 22 (msg:"LAB SSH Connection Attempt"; flags:S; sid:1000002; rev:1;)
```

The first rule demonstrates detection of ICMP echo requests. The second demonstrates an alert for TCP connection attempts toward SSH in the authorized lab. Rules should be tuned to reduce false positives.

6. Alert Monitoring

Suricata can generate structured events such as timestamps, source/destination addresses, ports, protocol information, rule identifiers, and alert messages. Analysts can review these events to determine whether activity is benign, suspicious, or requires escalation.

```
# Example: inspect recent alert records
# (path may vary by installation)
tail -f /var/log/suricata/fast.log

# Structured event log
tail -f /var/log/suricata/eve.json
```

7. Detection Workflow

1. Capture authorized network traffic.
2. Decode traffic by protocol.
3. Compare traffic against configured detection rules.
4. Generate an alert when a rule matches.
5. Record the event in logs.
6. Analyst reviews source, destination, time, protocol, and rule context.
7. Classify the event and take an appropriate response.

8. Response Mechanisms

An IDS normally detects and reports; response can be performed by an analyst or by carefully controlled automation. A safe response workflow is:

- Validate the alert and check for false positives.
- Record the event and preserve relevant logs.
- Identify the affected host or account where authorized.
- Isolate or block the source only when justified and permitted by the network owner.
- Investigate related events.
- Remediate the affected system where necessary.
- Monitor for recurrence and document the outcome.

9. Sample Alert Investigation

Field	Example
Timestamp	2026-10-08 12:00:00
Alert	LAB SSH Connection Attempt
Source	192.168.1.50
Destination	192.168.1.10:22
Protocol	TCP
Action	Review → validate → investigate → respond if authorized

10. Optional Visualization

Alert data can be visualized using a dashboard or graphing system. Useful charts include alerts over time, top source addresses, top destination ports, alert categories, and severity distribution. Visualization should support investigation rather than replace analyst verification.

11. Testing Plan

- Verify Suricata configuration before starting.
- Confirm the authorized interface is monitored.
- Generate benign test traffic in the lab.
- Confirm the expected rule triggers.
- Check fast.log and EVE JSON for the alert.
- Verify timestamp, source, destination, protocol, and rule ID.
- Tune rules if legitimate traffic creates excessive alerts.
- Document the final configuration and test results.

12. Security and Ethics

The IDS must only monitor traffic that the operator is authorized to inspect. Testing should be performed in a lab or on systems where explicit permission has been obtained. Logs may contain sensitive network information, so access to them should be restricted and retention should follow the organization's security policy.

13. Expected Outcome

After completion, the lab should be able to monitor authorized network traffic, detect selected suspicious patterns, generate alerts, retain useful event information, and support a documented investigation and response workflow.

14. Conclusion

This project demonstrates the core lifecycle of a network IDS: monitoring, detection, alerting, investigation, and response. Suricata rules provide a practical way to identify defined traffic patterns, while structured logs provide evidence for analysis. Proper tuning, authorization, and documented response procedures are essential for an effective IDS.

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